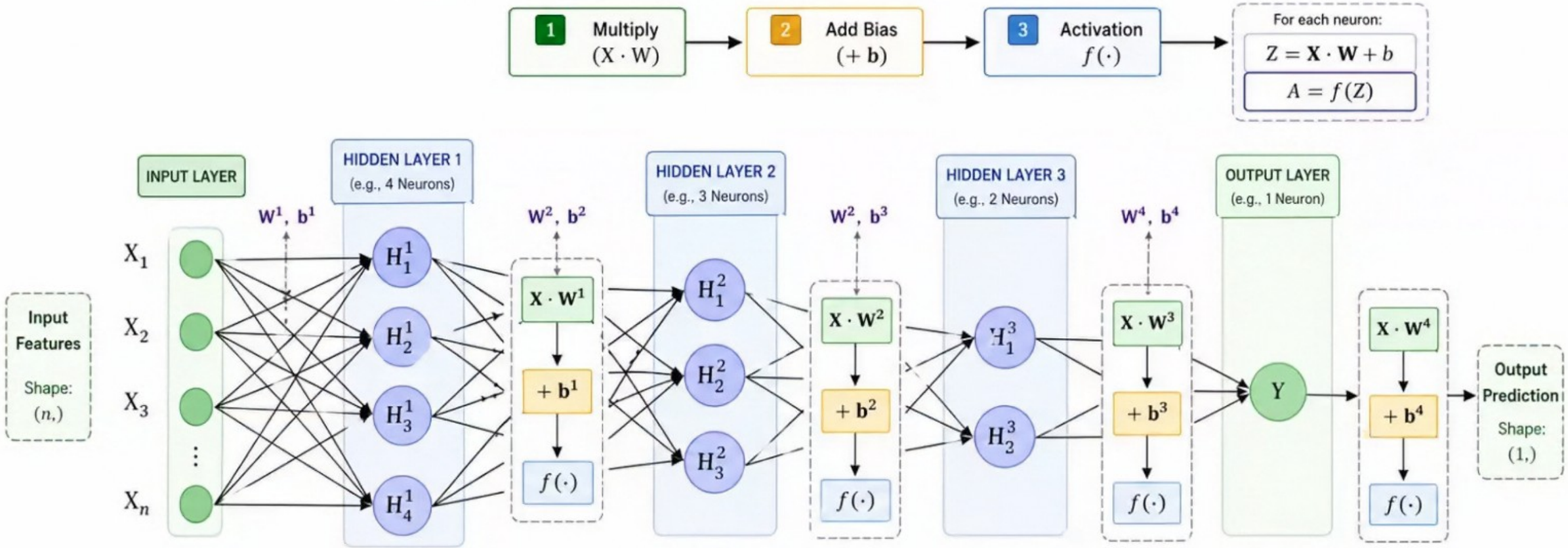
Artificial Neural Network (ANN) – Pipeline



• X = Input vector • n = number of input features	• W^1, b^1 = Weights and Bias for Hidden Layer 1	• W^2, b^2 = Weights and Bias for Hidden Layer 2	• W^3, b^3 = Weights and Bias for Hidden Layer 3	• W^4, b^4 = Weights and Bias for Output Layer

FLOW SUMMARY

Inputs (X) → Multiply with Weights $(X \cdot W)$ → Add Bias $(+ b)$ → Apply Activation Function $f(\cdot)$ → Next Layer → ⋯ → Output / Prediction

COMMON ACTIVATION FUNCTIONS

- ReLU: $f(z) = \max(0, z)$
- Sigmoid: $f(z) = 1/(1 + e^{-z})$
- Tanh: $f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$